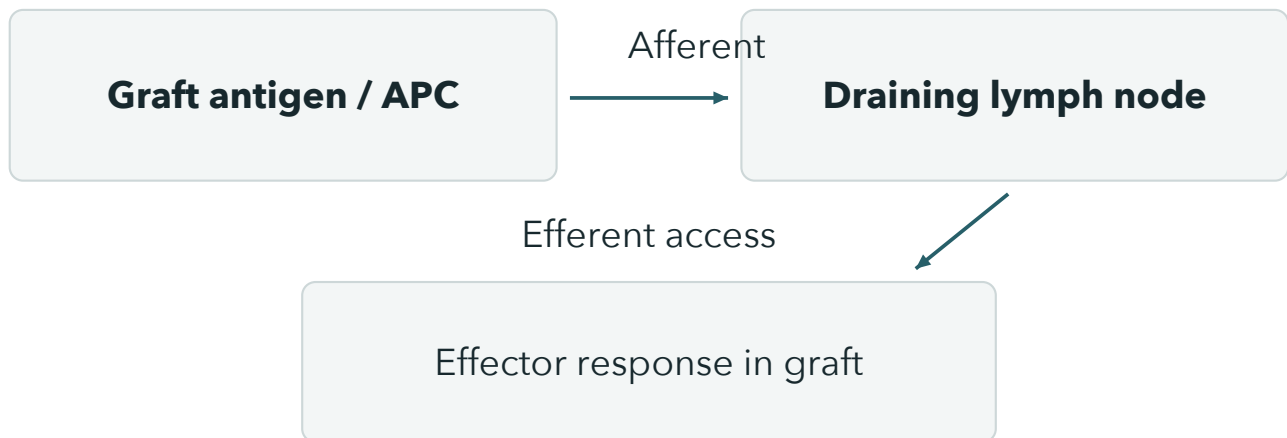


Immune privilege and graft rejection

Part 1 | Lymphatic access strengthens the afferent immune limb

Blood vessels and lymphatic vessels contribute different routes to graft immunity.



Antigen drainage and antigen-presenting-cell movement toward a lymph node precede the effector return to the graft.

A mechanistic distinction

Corneal lymphangiogenesis can facilitate antigen transport and immune priming in draining nodes. Blood vessels can support delivery of circulating effectors. Calling a vascularised cornea merely better nourished misses the separate immunological consequences of afferent and efferent access.

Priming and tissue injury are different stages

An intervention can affect the initiation of a response without directly killing its effector cells.

Stage

Feature	Main event	Possible experimental target
Afferent limb	Antigen/APC reaches node	Drainage or migration
Priming	Antigen-specific activation	Node interaction
Efferent limb	Effectors reach graft	Recruitment and local response

Experimental interpretation

If lymphatic inhibition reduces sensitisation while blood-vessel density and direct effector exposure are matched, the afferent route is a plausible target. Such a result does not establish that lymphatic blockade is a complete replacement for clinical immunosuppression.

Immune privilege is active and conditional

A privileged site can shape responses rather than simply hide every antigen.

Experimental ACAID concept

- Antigen enters anterior chamber
↓
- Systemic antigen handling occurs
↓
- Regulatory immune deviation develops
↓
- Selected inflammatory responses are reduced

The important limit

ACAID is an experimentally characterised antigen-specific pattern, not a claim that the patient has no systemic immunity. Preserved antibody responses can coexist with reduced delayed-type hypersensitivity. State which immune readout changed before concluding that all immunity has been suppressed.

A selective response needs a selective conclusion

One suppressed immune endpoint does not establish universal tolerance.

Observed finding

Feature	Supported interpretation	Overclaim
Reduced antigen-specific DTH	Selected cellular response altered	All infection defence absent
Preserved antibody response	Humoral function remains detectable	No immune deviation occurred
Loss of effect after pathway interruption	Pathway contributes in model	Every human response is identical

Study design

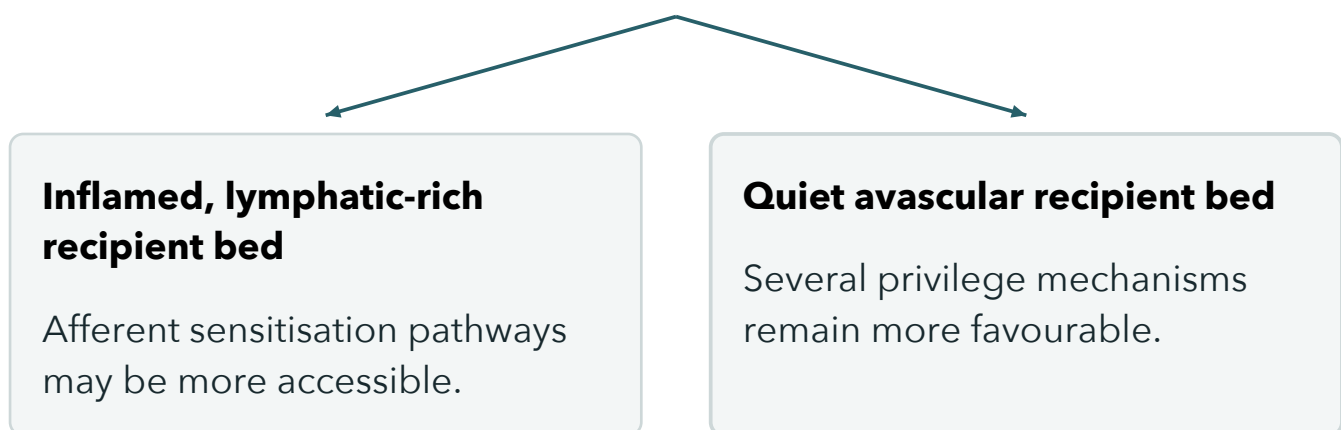
Compare the same antigen across readouts and include an unrelated antigen control. This separates antigen-specific deviation from general sickness, drug toxicity or a globally impaired immune system. The specificity of the conclusion should match the specificity of the experiment.

Translate mechanisms without erasing clinical context

A graft's risk reflects tissue history as well as intrinsic privilege.

Two otherwise similar graft beds

What differs?



Postgraduate interpretation

Separate mechanistic evidence from treatment claims. A successful murine pathway manipulation may identify a causal contributor; it does not automatically supply a human dose, schedule or clinical substitute. In an examination, use the demonstrated route to predict the affected immune stage.